

Enabling Intelligent Multi-AP Coordination for Wi-Fi 8

Deep Sudhanvabhai Gajjar, and Pradyumna Kumar Bishoyi, *Member, IEEE*,

Abstract—Coordinated spatial reuse (C-SR) is one of the multi-access point coordination (MAPC) modes of IEEE 802.11bn (Wi-Fi 8). In C-SR, the access point (AP) that wins a transmission opportunity (TXOP) shares it with a second AP, and a controller has to select the AP and the station that join the TXOP. We model this selection with a two-level hierarchical multi-armed bandit (H-MAB) and evaluate ϵ -greedy and upper confidence bound (UCB) agents. We then add a weighted per-station urgency term to the selection index, which gives urgency-aware (UA) versions of both agents. In a four-AP deployment, UCB reaches 98.8% of the oracle data rate, compared with 96.8% for ϵ -greedy, and obtains half of its final gain in 0.44 s instead of 2.92 s. With urgency, UA-UCB attains 97.7% of the oracle utility and serves more urgent stations than UA- ϵ -greedy at the same data rate.

Index Terms—IEEE 802.11bn, Wi-Fi 8, multi-AP coordination, coordinated spatial reuse, multi-armed bandits.

I. INTRODUCTION

IN recent years, dense Wi-Fi deployments have made co-channel interference between overlapping basic service sets a major limit on throughput and latency. IEEE 802.11bn (Wi-Fi 8) therefore introduces multi-AP coordination (MAPC) [1], [2]. In coordinated spatial reuse (C-SR), the AP that wins a TXOP, called the sharing AP, invites a shared AP to transmit at the same time on the same channel [3], [4].

A. Motivation

The gain of C-SR depends on which AP-station pair is added to the TXOP. Selection based on received signal strength (RSS) reports suffers from RSS variance and signaling overhead [5], [6]. Multi-armed bandits (MABs) avoid this exchange by learning from the outcome of past transmissions [4], [7]. Existing MAB schemes maximize the data rate only, whereas Wi-Fi 8 also targets low latency [1], so station urgency should be part of the selection.

B. Literature Survey

C-SR group selection has been studied with link rate estimation and group scheduling [3], [8], RSSI-based SINR prediction [9], testbed controllers [10], [11], analytical throughput models [12] and power control [13], [14]. A survey of MAPC is given in [15]. MABs were earlier applied to uncoordinated spatial reuse [16]–[20]. Wojnar *et al.* proposed the H-MAB

for C-SR and found UCB to perform best [4], and later added transmit power selection and testbed validation [7]. In these works the reward is the aggregate data rate.

C. Contributions

- 1) A formulation of two-AP C-SR selection with the sharing pair as context, with oracle bounds for the data rate and for a rate-urgency utility.
- 2) A comparison of ϵ -greedy and UCB agents in the H-MAB, including the steady-state accuracy limit of ϵ -greedy.
- 3) Urgency-aware ϵ -greedy and UCB agents that add a weighted urgency term to the selection index and learn only from the data rate.

II. SYSTEM MODEL

A. Network and Channel Access

We consider the downlink of N APs, $\mathcal{A} = \{A_1, \dots, A_N\}$, connected to a central controller [4]. AP A_i serves $\mathcal{S}_i = \{S_i^1, \dots, S_i^{m_i}\}$, and $M = \sum_{i=1}^N m_i$. Node X is at $\mathbf{p}_X \in \mathbb{R}^2$, and $\delta_{XY} = \|\mathbf{p}_X - \mathbf{p}_Y\|_2$. All stations are backlogged.

Time is divided into TXOPs of duration τ , indexed by t . In TXOP t , the sharing AP $A_k(t)$ is drawn uniformly from $\mathcal{A}$ and its station $S_k^l(t)$ uniformly from $\mathcal{S}_k$, which models DCF contention [4]. The sharing pair is $P_0(t) = (A_k(t), S_k^l(t))$. The controller selects a shared pair $P_1(t) = (A_i(t), S_i^j(t))$ from

$$\mathcal{X}(A_k) = \left\{ (A_i, S_i^j) : A_i \in \mathcal{A} \setminus \{A_k\}, S_i^j \in \mathcal{S}_i \right\}, \quad (1)$$

with $|\mathcal{X}(A_k)| = M - m_k$, and both pairs transmit concurrently.

B. Propagation and Link Model

The path loss follows the TGax enterprise model [4], [21]:

$$\text{PL}(\delta) = L_0 + 20 \log_{10} \bar{\delta} + \mathbf{1}\{\delta > B_p\} 35 \log_{10} \frac{\delta}{B_p} + 7W, \quad (2)$$

where $\bar{\delta} = \min\{\max\{\delta, 1\}, B_p\}$, B_p is the breakpoint distance, W is the number of walls and $L_0 = 40.05 + 20 \log_{10}(f_c/2.4)$ for the carrier frequency f_c in GHz. The received power is $P_{XY} = P_{\text{tx}} - \text{PL}(\delta_{XY})$. For the shared pair (A_i, S) , the SINR at S is

$$\gamma_t = P_{A_i S} - 10 \log_{10} \left(10^{P_{A_k S}/10} + 10^{N_0/10} \right) + \varepsilon_t, \quad (3)$$

D. S. Gajjar is with the Department of Electrical Engineering, Indian Institute of Technology Jodhpur, Rajasthan, India, 342030 (email: b23ee1014@iitj.ac.in).

P. K. Bishoyi is with the Department of Electronics Engineering, Indian Institute of Technology Jodhpur, Rajasthan, India, 342030, (email: pradyumna@iitj.ac.in).

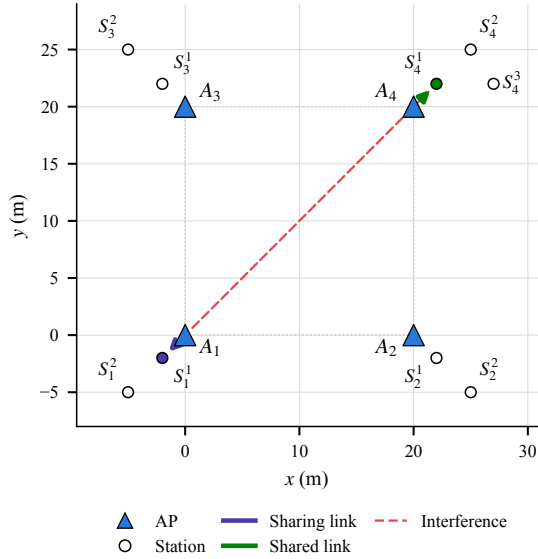


Figure 1: Simulated deployment. The arrows show one C-SR configuration with $P_0 = (A_1, S_1^1)$ and $P_1 = (A_4, S_4^1)$.

with noise floor N_0 and $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$ [4]. The delivered rate is

$$r_t = R_{\max} \eta(\gamma_t), \quad (4)$$

$$\eta(\gamma) = \min \left\{ 1, \max \left\{ \eta_{\min}, \frac{\gamma}{\gamma_{\text{ref}}} \right\} \right\}. \quad (5)$$

Only the interference of A_k at S is modeled, so r_t is the rate added by C-SR. The expected rate of a shared pair is

$$\mu(A_k, A_i, S) = \mathbb{E}_\varepsilon[r_t] = R_{\max} \int_{-\infty}^{\infty} \eta(\tilde{\gamma} + x) \phi_\sigma(x) dx, \quad (6)$$

where $\tilde{\gamma}$ is (3) with $\varepsilon_t = 0$ and ϕ_σ is the $\mathcal{N}(0, \sigma^2)$ density.

C. Station Urgency

Each station S has an urgency $u_S(t) \in [0, 1]$, known at the start of TXOP t and constant during it. In this letter, $u_S(t)$ is drawn independently from the uniform distribution on $[0, 1]$ in every TXOP.

D. Problem Formulation

The controller does not know μ and observes only r_t . The throughput objective is

$$\bar{r} = \lim_{T \rightarrow \infty} \frac{1}{T} \sum_{t=1}^T \mathbb{E}[r_t]. \quad (7)$$

With known μ , the optimal choice and the oracle rate are

$$P_1^*(A_k) = \arg \max_{(A_i, S) \in \mathcal{X}(A_k)} \mu(A_k, A_i, S), \quad (8)$$

$$\bar{r}^* = \frac{1}{N} \sum_{k=1}^N \mu(A_k, P_1^*(A_k)), \quad (9)$$

and the regret of a policy is

$$\rho(T) = \sum_{t=1}^T \left[\mu(A_k(t), P_1^*(A_k(t))) - \mu(A_k(t), P_1(t)) \right]. \quad (10)$$

With urgency, the per-TXOP utility is $U_t = r_t + w u_{S_1(t)}(t)$, where $S_1(t)$ is the station of $P_1(t)$ and $w \geq 0$ is in Mb/s. The utility oracle selects

$$P_1^w(t) = \arg \max_{(A_i, S) \in \mathcal{X}(A_k)} [\mu(A_k, A_i, S) + w u_S(t)]. \quad (11)$$

III. PROPOSED SOLUTION

A. Hierarchical MAB

Following [4], the selection is split into two levels.

1) *State space*: The context is $P_0 = (A_k, S_k^l)$. There are M first-level agents $\alpha^I[P_0]$ and $(N-1)M$ second-level agents $\alpha^{II}[P_0, A_i]$.

2) *Actions*: $\alpha^I[P_0]$ selects $A_i \in \mathcal{A} \setminus \{A_k\}$ ($N-1$ arms), and $\alpha^{II}[P_0, A_i]$ selects $S_i^j \in \mathcal{S}_i$ (m_i arms), which gives $P_1 = (A_i, S_i^j) \in \mathcal{X}(A_k)$.

3) *Reward function*: Both agents receive r_t . For each arm x , an agent keeps $Q(x)$ and $n(x)$, initialized to zero, and updates

$$Q(x) \leftarrow Q(x) + \alpha [r_t - Q(x)], \quad n(x) \leftarrow n(x) + 1, \quad (12)$$

with $\alpha = 1/2$. After n updates,

$$Q_n(x) = \sum_{\nu=1}^n \alpha (1-\alpha)^{n-\nu} r^{(\nu)}, \quad \mathbb{E}[Q_n] = (1 - (1-\alpha)^n) \mu, \quad (13)$$

and for i.i.d. rewards with variance σ_r^2 , $\text{Var}[Q_n] \rightarrow \alpha \sigma_r^2 / (2 - \alpha) = \sigma_r^2 / 3$ [22].

B. Exploration Strategies

1) *ϵ -greedy*: A random arm is chosen with probability ϵ , otherwise $\arg \max_x Q(x)$. When both levels rank their arms correctly, $P_1^*(A_k)$ is chosen with probability

$$\pi_\epsilon(A_k) = \left(1 - \epsilon + \frac{\epsilon}{N-1} \right) \left(1 - \epsilon + \frac{\epsilon}{m_{i^*}} \right), \quad (14)$$

where A_{i^*} is the AP in $P_1^*(A_k)$.

2) *UCB*: Each arm is played once, and then

$$x_t = \arg \max_x \left[Q(x) + c \sqrt{\frac{\ln n_{\text{tot}}}{n(x)}} \right], \quad (15)$$

where $n_{\text{tot}} = \sum_{x'} n(x')$ and c is in Mb/s. Since Q follows (13) instead of a sample mean, the regret bound of UCB1 [23] does not apply directly, and c is tuned empirically.

C. Urgency-Aware H-MAB

In the UA schemes, the sharing AP serves the station of S_k with the largest $u_S(t)$, and each agent selects the arm with the largest index

$$I^I(A_i) = Q_{P_0}^I(A_i) + w \max_{S \in \mathcal{S}_i} u_S(t) + b^I(A_i), \quad (16)$$

$$I^{II}(S) = Q_{P_0, A_i}^{II}(S) + w u_S(t) + b^{II}(S), \quad (17)$$

where $b(x) = 0$ for UA- ϵ -greedy, which keeps its random choices with probability ϵ , and $b(x) = c \sqrt{\ln n_{\text{tot}} / n(x)}$ for UA-UCB. The update (12) is unchanged, so Q estimates only the rate. Since $u_S(t) \in [0, 1]$, urgency can reverse the order of two arms with $Q(x) > Q(x')$ only if

$$w > Q(x) - Q(x'). \quad (18)$$

Algorithm 1: H-MAB With Urgency-Aware Selection

Inputs: $\mathcal{A}$, $\{\mathcal{S}_i\}_{i=1}^N$, rule (ϵ -greedy or UCB), ϵ or c , w
 Set $Q(x) = 0$, $n(x) = 0$ for all arms;
for each TXOP t do
 Draw A_k uniformly from $\mathcal{A}$ and observe $\mathbf{u}(t)$;
 if $w > 0$ **then**
 $S_k^l \leftarrow \arg \max_{S \in \mathcal{S}_k} u_S(t)$;
 else
 Draw S_k^l uniformly from $\mathcal{S}_k$;
 $P_0 \leftarrow (A_k, S_k^l)$;
 $A_i \leftarrow$ arm of $\alpha^I[P_0]$ with the largest index (16);
 $S_i^j \leftarrow$ arm of $\alpha^{II}[P_0, A_i]$ with the largest index (17);
 Transmit $A_k \rightarrow S_k^l$ and $A_i \rightarrow S_i^j$, and observe r_t ;
 Update $\alpha^{II}[P_0, A_i]$ and $\alpha^I[P_0]$ with (12);

For $w \rightarrow \infty$, $S_1(t)$ is the most urgent candidate, and

$$\mathbb{E}[\bar{u}_1] \rightarrow \frac{1}{N} \sum_{k=1}^N \frac{M - m_k}{M - m_k + 1}. \quad (19)$$

Algorithm 1 summarizes all schemes; $w = 0$ gives the urgency-agnostic agents.

D. Complexity

Each TXOP needs $O(N + \max_i m_i)$ operations. The controller stores $2[M(N-1) + \sum_{k=1}^N m_k(M-m_k)]$ values, which is 174 for the deployment of Fig. 1.

IV. NUMERICAL RESULTS

A. Simulation Setup

The deployment is shown in Fig. 1 and the parameters in Table I. Stations lie on the outward diagonal at 2.83 m and 7.07 m from their AP, and S_4^3 is at 7.28 m. Each run has $T = 6000$ TXOPs (32.9 s), all schemes use the same sharing APs and urgencies, and results are averaged over 30 runs. ϵ and c were tuned on 30 separate runs by maximizing the run average of r_t (and of U_t for the UA schemes), which gave $\epsilon = 0.1$ and $c = 18$ Mb/s in both cases.

We report the mean of r_t (or U_t) over the last 1000 TXOPs, $\bar{r}$ (or $\bar{U}$), and over the whole run, $\bar{r}_{\text{run}}$. t_{50} and t_{90} are the times at which the 50-TXOP moving average covers 50% and 90% of its gain from TXOP 50 to the steady state. π^* and π^w are the fractions of the last 1000 TXOPs in which $P_1(t)$ equals (8) and (11), respectively. μ in (6) is computed with 4×10^5 samples.

For every A_k , $P_1^*(A_k)$ is the diagonal AP with its near station, with a mean SINR of 28.2 dB and $\mu = 96.7$ Mb/s. The near station of an adjacent AP gives 23.0 dB and 78.9 Mb/s, so the smallest gap to the best pair is 17.8 Mb/s. Hence, $\bar{r}^* = 96.73$ Mb/s, and random selection gives 73.81 Mb/s.

B. Urgency-Agnostic Schemes

Fig. 2 and Table II compare random selection, ϵ -greedy and UCB. UCB reaches t_{50} in 0.44 s and $\bar{r} = 95.53$ Mb/s, which

Table I: Simulation Parameters

Parameter	Value
N, m_i, M	4; 2, 2, 2, 3; 9
AP positions	corners of a 20×20 m square
f_c, B_p, W	5 GHz, 10 m, 0
P_{tx}, N_0	16.02 dBm, -93.97 dBm
σ	2 dB
$R_{\text{max}}, \gamma_{\text{ref}}, \eta_{\text{min}}$	120 Mb/s, 35 dB, 0.1
τ	5.484 ms [24]
T , runs	6000 TXOPs, 30
α, ϵ, c	1/2, 0.1, 18 Mb/s
w	60 Mb/s (0 to 300 Mb/s in Fig. 3(b))

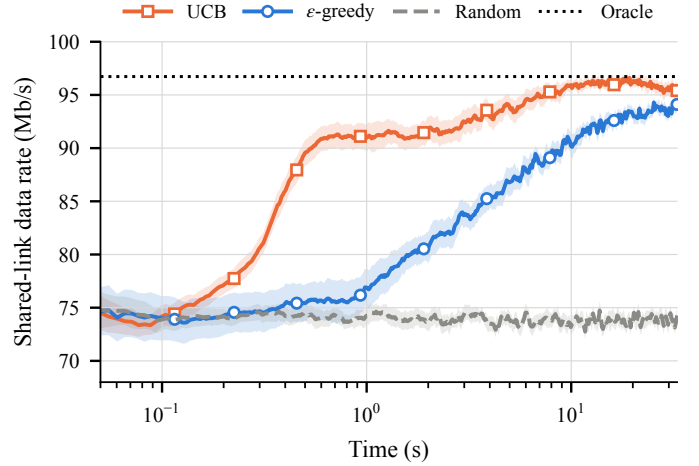


Figure 2: Shared-link data rate of the urgency-agnostic schemes (30 runs, 50-TXOP moving average, 95% confidence bands).

is 98.8% of $\bar{r}^*$. ϵ -greedy needs 2.92 s and reaches 93.66 Mb/s. Its $\pi^* = 86.5\%$ is close to the limit of 88.3% given by (14).

Table III shows that $\epsilon = 0$ does not learn and that UCB with $c \leq 12$ Mb/s loses about 4 Mb/s. With $c = 0$, 81 of the 270 first-level agents lock onto an adjacent AP, in 75 cases because the first trial of the diagonal AP went to a far station. With $c = 18$ Mb/s, 18 agents remain locked.

C. Urgency-Aware Schemes

The UA schemes are compared with the most-urgent-first rule, which selects the most urgent candidate station without learning, and with the utility oracle (11). For $w = 60$ Mb/s (Fig. 3(a) and Table IV), UA-UCB attains $\bar{U} = 130.90$ Mb/s (97.7% of $\bar{U}^w$) and UA- ϵ -greedy 128.14 Mb/s (95.6%). The most-urgent-first rule gives 125.84 Mb/s with $\bar{u}_1 = 0.87$ but only 73.55 Mb/s of rate. UA-UCB passes this rule after 0.57 s and UA- ϵ -greedy after 5.89 s. The initial drop of UA- ϵ -greedy occurs because tried arms, with $Q \approx r/2$ from (13), outrank untried arms regardless of urgency.

Fig. 3(b) shows the rate against $\bar{u}_1$ for w from 0 to 300 Mb/s. For every $w \geq 10$ Mb/s, UA-UCB gives both a higher rate and a higher $\bar{u}_1$ than UA- ϵ -greedy and stays closer to the oracle curve. In agreement with (18), $w \leq 10$ Mb/s, which is below the 17.8 Mb/s gap, has almost no effect. At $w = 300$ Mb/s, UA-UCB reaches $\bar{u}_1 = 0.866$, close to the limit of 0.87 from (19), while UA- ϵ -greedy stops at 0.825 because of its random choices.

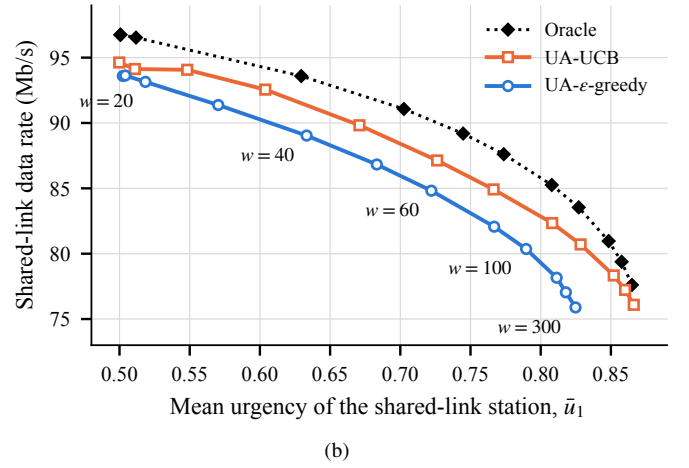
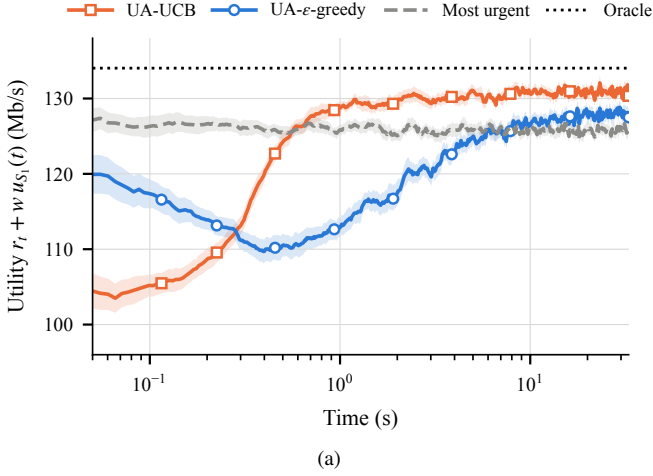


Figure 3: Urgency-aware schemes (30 runs). (a) Utility for $w = 60$ Mb/s (50-TXOP moving average, 95% confidence bands). (b) Steady-state rate versus $\bar{u}_1$. The markers on each curve correspond to $w = 0, 10, 20, 30, 40, 50, 60, 80, 100, 150, 200$ and 300 Mb/s from left to right.

Table II: Urgency-Agnostic Schemes (95% Confidence Half-Width of $\bar{r}$ at Most 0.5 Mb/s)

Scheme	$\bar{r}$ (Mb/s)	$\bar{r}/\bar{r}^*$	$\bar{r}_{\text{run}}$ (Mb/s)	t_{50} (s)	t_{90} (s)	π^*
Random	73.85	76.3%	73.85	n/a	n/a	15.4%
ϵ -greedy	93.66	96.8%	90.69	2.92	11.42	86.5%
UCB	95.53	98.8%	95.11	0.44	4.69	93.0%
Oracle	96.73	100%	96.73	n/a	n/a	100%

Table IV: Urgency-Aware Schemes, $w = 60$ Mb/s (95% Confidence Half-Width of $\bar{U}$ and $\bar{r}$ at Most 0.21 Mb/s)

Scheme	$\bar{U}$ (Mb/s)	$\bar{U}/\bar{U}^w$	$\bar{r}$ (Mb/s)	$\bar{u}_1$	t_{50} (s)	t_{90} (s)	π^w
Most urgent	125.84	93.9%	73.55	0.87	n/a	n/a	46.4%
UA- ϵ -greedy	128.14	95.6%	84.82	0.72	2.25	8.06	63.1%
UA-UCB	130.90	97.7%	84.92	0.77	0.43	1.03	69.3%
Oracle	134.03	100%	87.61	0.77	n/a	n/a	100%

Table III: Sensitivity of the Urgency-Agnostic Schemes (95% Confidence Half-Width of $\bar{r}$ at Most 1.5 Mb/s)

ϵ	$\bar{r}$	$\bar{r}_{\text{run}}$	π^*	c	$\bar{r}$	$\bar{r}_{\text{run}}$	π^*
0	75.02	74.94	20.9%	0	90.99	90.84	68.1%
0.02	90.88	86.18	70.0%	6	91.07	90.91	68.5%
0.05	93.74	89.67	86.2%	12	92.69	91.77	77.0%
0.1	93.66	90.69	86.5%	18	95.53	95.11	93.0%
0.2	91.52	90.00	74.7%	24	95.23	92.18	92.1%
0.3	89.29	88.39	64.6%	36	95.11	93.02	92.4%
				48	94.57	91.46	89.7%
				96	90.66	86.86	71.9%

V. CONCLUSION

In this letter, we modeled two-AP C-SR in IEEE 802.11bn as a contextual selection problem and solved it with an H-MAB that learns from the observed data rate. UCB outperformed ϵ -greedy in learning speed and steady-state rate and reached 98.8% of the oracle rate. Adding a weighted urgency term to the selection index gave UA agents whose trade-off is set by a single weight, and UA-UCB reached 97.7% of the oracle utility. Future work will include the sharing-link loss in the reward, larger C-SR groups and queue-based urgency.

REFERENCES

- [1] L. Galati-Giordano, G. Geraci, M. Carrascosa, and B. Bellalta, "What will Wi-Fi 8 be? A primer on IEEE 802.11bn ultra high reliability," *IEEE Commun. Mag.*, vol. 62, no. 8, pp. 126–132, Aug. 2024.
- [2] R. J. Yu, "Specification framework for TGBn," IEEE 802.11 document 802.11-24/0209r6, Nov. 2024.
- [3] D. Nunez, F. Wilhelmi, S. Avallone, M. Smith, and B. Bellalta, "TXOP sharing with coordinated spatial reuse in multi-AP cooperative IEEE 802.11be WLANs," in *Proc. IEEE 19th Annu. Consum. Commun. Netw. Conf. (CCNC)*, Jan. 2022, pp. 864–870.
- [4] M. Wojnar, W. Ciezobka, K. Kosek-Szott, K. Rusek, S. Szott, D. Nunez, and B. Bellalta, "IEEE 802.11bn multi-AP coordinated spatial reuse with hierarchical multi-armed bandits," *IEEE Commun. Lett.*, vol. 29, no. 3, pp. 428–432, Mar. 2025.
- [5] Y. Kim, H. Shin, Y. Chon, and H. Cha, "Smartphone-based Wi-Fi tracking system exploiting the RSS peak to overcome the RSS variance problem," *Pervasive Mobile Comput.*, vol. 9, no. 3, pp. 406–420, Jun. 2013.
- [6] Y. Seok, "Coordinated spatial reuse (C-SR) protocol," IEEE 802.11 document 802.11-20/0576r1, May 2020.
- [7] M. Wojnar, W. Ciezobka, A. Tomaszewski, P. Cholda, K. Rusek, K. Kosek-Szott, J. Haxhibeqiri, J. Hoebeke, B. Bellalta, A. Zubow, F. Dressler, and S. Szott, "Coordinated spatial reuse scheduling with machine learning in IEEE 802.11 MAPC networks," *IEEE J. Sel. Areas Commun.*, vol. 43, no. 11, pp. 3666–3682, Nov. 2025.
- [8] D. Nunez, M. Smith, and B. Bellalta, "Multi-AP coordinated spatial reuse for Wi-Fi 8: Group creation and scheduling," arXiv:2305.04846, 2023.
- [9] P. Imputato, S. Avallone, M. Smith, D. Nunez, and B. Bellalta, "Beyond Wi-Fi 7: Spatial reuse through multi-AP coordination," *Comput. Netw.*, vol. 239, Feb. 2024, art. no. 110160.
- [10] J. Haxhibeqiri *et al.*, "Coordinated spatial reuse for WiFi networks: A centralized approach," in *Proc. IEEE 20th Int. Conf. Factory Commun. Syst. (WFCS)*, Apr. 2024, pp. 1–8.
- [11] J. Haxhibeqiri *et al.*, "Coordinated SR and restricted TWT for time sensitive applications in WiFi 7 networks," *IEEE Commun. Mag.*, vol. 62, no. 8, pp. 118–124, Aug. 2024.
- [12] F. Wilhelmi, L. Galati-Giordano, G. Geraci, B. Bellalta, G. Fontanesi, and D. Nunez, "Throughput analysis of IEEE 802.11bn coordinated spatial reuse," arXiv:2309.09169, 2023.
- [13] J. Wang and X. Fang, "Research on next-generation Wi-Fi spatial reuse power control based on federated reinforcement learning," in *Proc. IEEE 99th Veh. Technol. Conf. (VTC-Spring)*, Jun. 2024, pp. 1–6.
- [14] M. Talukder and J. Xie, "Enhanced coordinated spatial reuse: Bidirectional multiple AP coordination for IEEE 802.11be," in *Proc. IEEE Int. Conf. Commun. (ICC)*, May 2023, pp. 660–665.
- [15] S. Verma, T. K. Rodrigues, Y. Kawamoto, M. M. Fouda, and N. Kato, "A survey on multi-AP coordination approaches over emerging WLANs: Future directions and open challenges," *IEEE Commun. Surveys Tuts.*, vol. 26, no. 2, pp. 858–889, 2024.

- [16] A. Bardou, T. Begin, and A. Busson, “Improving the spatial reuse in IEEE 802.11ax WLANs: A multi-armed bandit approach,” in *Proc. 24th Int. ACM Conf. Model. Anal. Simul. Wireless Mobile Syst. (MSWiM)*, Nov. 2021, pp. 135–144.
- [17] F. Wilhelmi, S. Barrachina-Muñoz, B. Bellalta, C. Cano, A. Jonsson, and G. Neu, “Potential and pitfalls of multi-armed bandits for decentralized spatial reuse in WLANs,” *J. Netw. Comput. Appl.*, vol. 127, pp. 26–42, Feb. 2019.
- [18] F. Wilhelmi, C. Cano, G. Neu, B. Bellalta, A. Jonsson, and S. Barrachina-Muñoz, “Collaborative spatial reuse in wireless networks via selfish multi-armed bandits,” *Ad Hoc Netw.*, vol. 88, pp. 129–141, May 2019.
- [19] H. Kim, G. Na, H. Im, and J. So, “Improving spatial reuse of wireless LANs using contextual bandits,” *IEEE Trans. Wireless Commun.*, vol. 23, no. 7, pp. 6735–6749, Jul. 2024.
- [20] P. E. Iturria-Rivera, M. Chenier, B. Herscovici, B. Kantarci, and M. Erol-Kantarci, “Cooperate or not cooperate: Transfer learning with multi-armed bandit for spatial reuse in Wi-Fi,” *IEEE Trans. Mach. Learn. Commun. Netw.*, vol. 2, pp. 351–369, 2024.
- [21] S. Merlin *et al.*, “TGax simulation scenarios,” IEEE 802.11 document 802.11-14/0980r16, Nov. 2015.
- [22] R. S. Sutton and A. G. Barto, *Reinforcement Learning: An Introduction*, 2nd ed. Cambridge, MA, USA: MIT Press, 2018.
- [23] P. Auer, N. Cesa-Bianchi, and P. Fischer, “Finite-time analysis of the multiarmed bandit problem,” *Mach. Learn.*, vol. 47, no. 2–3, pp. 235–256, 2002.
- [24] K. Kosek-Szott and N. Rapacz, “Tuning Wi-Fi traffic differentiation by combining frame aggregation with TXOP limits,” *IEEE Commun. Lett.*, vol. 24, no. 3, pp. 700–703, Mar. 2020.